

Impact and Feature Analysis (SB5-MAI)

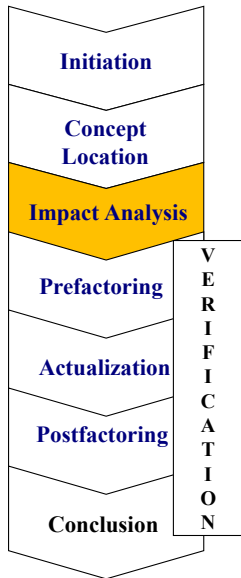
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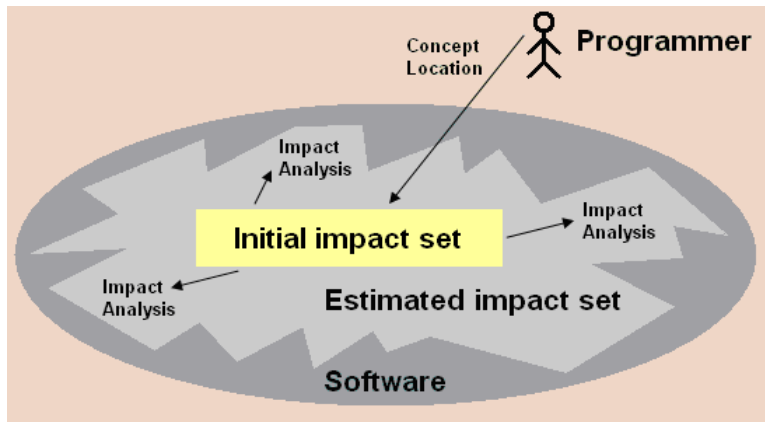
Impact and Feature Analysis

Impact analysis (IA)

- ▶ Determines the strategy and impact of change
- ▶ Classes identified in concept location are the initial impact set
- ▶ Class dependencies are analyzed, and impacted classes are added to the impact set
- ▶ Produces estimated impact set



Initial and estimated impact set



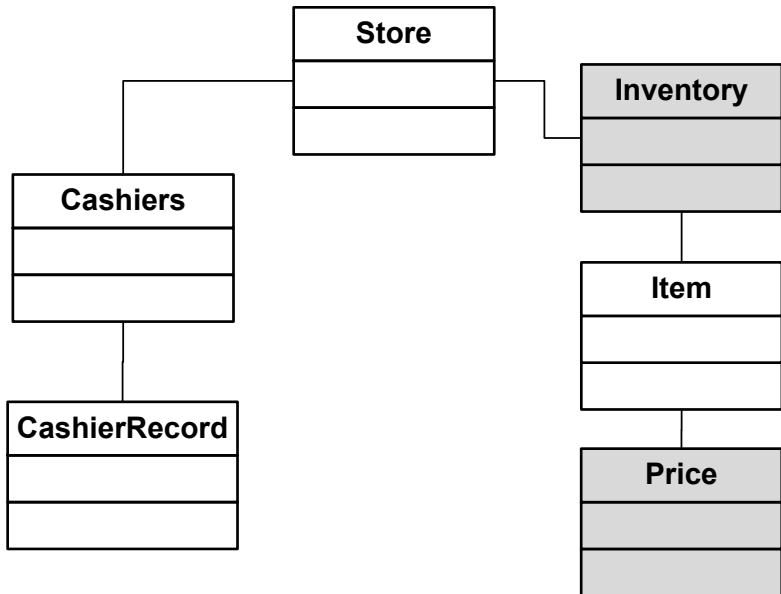
Class Interactions

- ▶ Two classes interact if they have something in common
 - ▶ One depends on the other. There is a contract between them
 - ▶ They coordinate. They share the same coding, schedule, etc.
- ▶ Interactions propagate change in both directions
 - ▶ From A to B or from B to A

Class Interaction Graph

- ▶ $G = (X, I)$
 - ▶ X ... set of classes
 - ▶ I ... set of interactions
- ▶ Neighborhood of class A
 - ▶ $N(A) = \{B \mid (A, B) \in I\}$

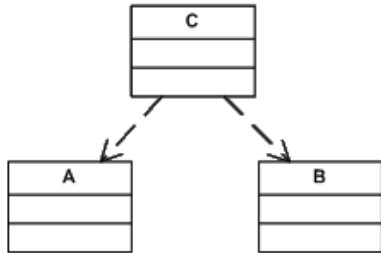
Neighborhood of Item



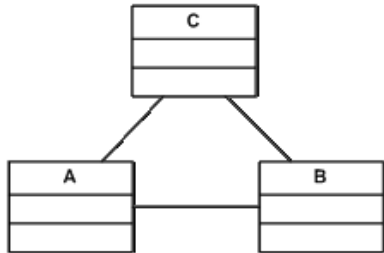
Coordination

```
class C
{
    A  a;          //gets the color code
    B  b;          //paints the screen
    void foo()
    {
        b.paint(a.get()); // dataflow between a and b
    }
};
```


Dependency diagram, Interaction diagram



Dependencies



Interactions

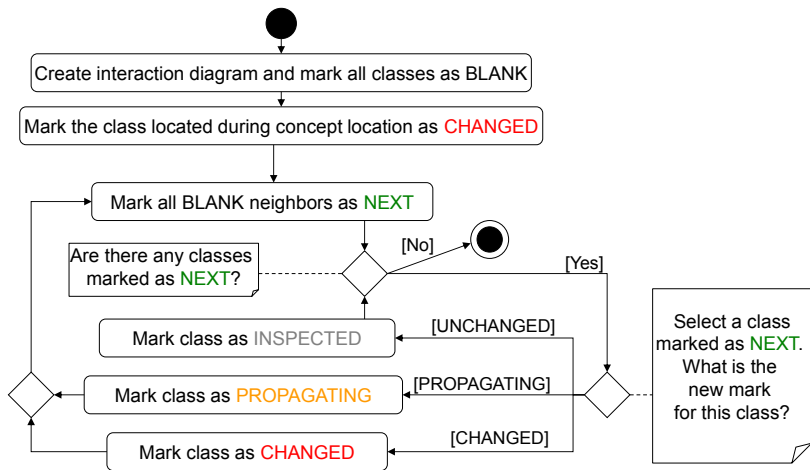
Propagating class: Mailman

- ▶ John loaned money to Paul:
 - ▶ Needs the money back
 - ▶ His situation changed
- ▶ John writes a letter to Paul:
 - ▶ Mailman takes the letter from John to Paul
 - ▶ Paul must take a part-time job
 - ▶ A big change that propagated from John to Paul
- ▶ John interacts with the mailman
- ▶ The mailman interact with Paul
- ▶ The change originated with John and propagates through the mailman to Paul

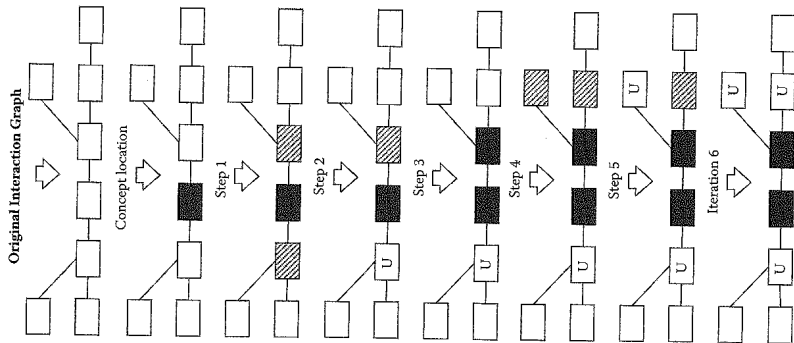
Process of Impact Analysis

Blank	The class was never inspected and is not scheduled for an inspection.
Changed	The programmers inspected the class and found that it is impacted by the change
Unchanged	The programmers inspected the class and found that it is not impacted by the change.
Next	The class is scheduled for inspection

IA including propagating classes



IA Example



Alternatives in Software Change

- ▶ Program displaying a temperature in Fahrenheit
 - ▶ change request: display it in Celsius
- ▶ Two separate locations deal with temperature
 - ▶ sensor data converted to the temperature
 - ▶ temperature displayed to the user
- ▶ The change can be done in either place
 - ▶ impact analysis weights these alternatives

The Criteria

- ▶ Required effort of the change
- ▶ Clarity of the resulting code
- ▶ Often, these two criteria contradict each other:
 - ▶ it is easier to adjust the user interface
 - ▶ it is better to have all calculations of the temperature in one place
- ▶ Conflict between short-term and long-term goals

Interactive IA

